

Experiment ‘p_4_6_find_minimum_seq’ Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_4_6_find_minimum_seq

Natural language statement: Write a method that finds the minimum value from a list of inputs into a method.

Method signature: p_4_6_find_minimum_seq(inputs: seq<int>) returns (min: int)

Ensures

- forall i :: 0 <= i < |inputs| ==> min <= inputs[i]
- exists i :: 0 <= i < |inputs| && min == inputs[i]

Requires

- 0 < |inputs|

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that finds the minimum value from a list of inputs into a method.

The signature should be:

`p_4_6_find_minimum_seq(inputs: seq<int>) returns (min: int)`

The method should respect the following contract:

requires $0 < |inputs|$, ensures forall $i :: 0 \leq i < |inputs| \implies \min \leq inputs[i]$,
 ↪ ensures exists $i :: 0 \leq i < |inputs| \ \&\& \ \min == inputs[i]$

Produce and show only the Dafny body of this method, including the curly braces that
 ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmfdHbRhGD2K0OC2aTN4E88Y02saQ

Model: gpt-5.5-2026-04-23

Created at: 1780492151

Finish reason: stop

Usage: 338 tokens in, and 693 tokens out

```
{
  min := inputs[0];
  var i := 1;
  var k := 0;

  while i < |inputs|
    invariant 1 <= i <= |inputs|
    invariant 0 <= k < i
    invariant min == inputs[k]
    invariant forall j :: 0 <= j < i ==> min <= inputs[j]
  {
    if inputs[i] < min {
      min := inputs[i];
      k := i;
    }
    i := i + 1;
  }

  assert 0 <= k < |inputs| && min == inputs[k];
  assert exists j :: 0 <= j < |inputs| && min == inputs[j];
}
```

```
}
```

Final Program

```
method p_4_6_find_minimum_seq(inputs: seq<int>) returns (min: int)
  requires 0 < |inputs|
  ensures forall i :: 0 <= i < |inputs| ==> min <= inputs[i]
  ensures exists i :: 0 <= i < |inputs| && min == inputs[i]
{
  min := inputs[0];
  var i := 1;
  var k := 0;

  while i < |inputs|
    invariant 1 <= i <= |inputs|
    invariant 0 <= k < i
    invariant min == inputs[k]
    invariant forall j :: 0 <= j < i ==> min <= inputs[j]
  {
    if inputs[i] < min {
      min := inputs[i];
      k := i;
    }
    i := i + 1;
  }

  assert 0 <= k < |inputs| && min == inputs[k];
  assert exists j :: 0 <= j < |inputs| && min == inputs[j];
}
```

Total Token Usage

Input tokens: 338

Output tokens: 693

Reasoning tokens: 512

Sum of 'total tokens': 1031

Experiment Timings

Overall Experiment started at 1780492150588, ended at 1780492171091, lasting 20503ms (20.50 seconds)

Iteration #1 started at 1780492150588, ended at 1780492171091, lasting 20503ms (20.50 seconds)